

MODELING TO PREDICT THE PATIENTS' POSTOPERATIVE WOMAC SCORE BY FEATURES ENGINEERING AND GRADIENT BOOST TREE

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Abstract

This research studies factors and creates a model to predict the patients' postoperative WOMAC score after total knee replacement. First, the influencing factors were found by feature engineering, using several techniques such as Generalized Linear Models, Support Vector Machines, Deep Learning, and Gradient Boost Trees. Afterwards, the model was created by the Gradient Boost Tree technique which groups different attributes from feature engineering. Models were compared to find the model with the best predictability. RapidMiner Studio software version 9.9 was used in this work. The results demonstrate that the model created by the Gradient Boost Tree technique with attributes originating from feature engineering on the Gradient Boost Tree performs most efficiently with root mean square error (RMSE), mean absolute deviation (MAD) and square error (SE) of 5.311 ± 0.538 , 3.550 ± 0.376 , and 28.472 ± 5.811 respectively.

Keywords: Deep Learning; Generalized Linear Model; Gradient Boost Tree; Knee Osteoarthritis; Support Vector Machine; WOMAC score

Introduction

Numerous countries around the world are experiencing the aging of their societies nowadays. Thailand's data shows that the number of people who had reached an age of over 60 years were 15.45%, 16.05%, 16.73% and 17.57% of its population each year between 2017 and 2020, respectively (Department of Older Persons, 2020). The Department of Older Persons predicts that the percentage of the elderly of Thailand will increase

to 20.1% of the population by the year 2032.

The most common musculoskeletal disorder that the elderly experience is knee osteoarthritis. According to information from the Department of Medical Services, the number of patients experiencing knee osteoarthritis in Thailand was about 6 million people in 2018 (Department of Medical Services, 2020), indicating that knee osteoarthritis is indeed one of the important causes

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for the elderly to become disabled. Osteoarthritis can have a profound impact on a person's physical health. For example, stiffness and difficulty moving the knee give patients many difficulties in their daily lives. There is currently no cure for knee osteoarthritis. However, a variety of medical and nonmedical treatments can help relieve the symptoms, and total knee replacement can be used with patients in critical condition.

Several techniques are available for the diagnosis of knee conditions, such as physical examinations, clinical histories, X-ray images and Magnetic Resonance Imaging (MRI). Some of the diagnostic techniques may be difficult to perform and require much time, thus some researchers have created questionnaires to help assess the extent of knee osteoarthritis in patients. The Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) is widely used in various contexts of research because of its ease of implementation. The WOMAC consists of 24 items under three headings: pain, stiffness, and physical function, as measured by the patient in the previous 48 hours. The WOMAC score can be used in several treatments, for instance, in symptom follow-up and estimation of the extent of the disease within the patient. However, actual usage of the WOMAC is left to the attending physician's judgment.

Currently, the world is experiencing a 'digital economy' that has several technologies that are playing increasing roles in medicine. One such technology is artificial intelligence (AI) used for diagnosis. For instance, Yuan *et al.* (2020) developed AI for early sepsis diagnosis in intensive care units with 82% accuracy. Another technology that is increasingly being used to classify, predict or evaluate a patient's disease is machine learning (ML). For example, Tiulpin *et al.* (2019) predicted the potential need for total knee replacement using the Gradient Boost Machine (GBM) and Convolutional Neural Network (CNN), with an area under curve (AUC) accuracy of 0.79 (0.78-0.81).

To summarize, we can see that AI and ML are playing increasing roles in medicine to predict, classify or evaluate a patient's disease. This research applies feature engineering to find the relevant factors by selection and generation of new attributes. The optimizing Gradient Boost Tree to predict a patient's postoperative WOMAC score after total knee replacement by these attributes is found by feature engineering.

Related Mathematics Background

Questionnaire

In previous years, for ease in following up the symptoms and diagnosis of osteoarthritis, several questionnaires to evaluate the symptoms of osteoarthritis have been created, such as the Western Ontario and McMaster Universities Arthritis Index (WOMAC), Knee Injury and Osteoarthritis Outcome Score (KOOS), International Knee Documentation Committee (IKDC), and Oxford Knee Score (OKS). The WOMAC was developed in 1982, is available in over 65 languages, and is a self-administered questionnaire that takes approximately 12 minutes to complete for evaluation of Hip and Knee Osteoarthritis, as well as in research studies. The WOMAC consists of 24 items divided into three subsections which are Pain (5 items), Stiffness (2 items), and Physical Function (17 items). The patient fills out the form during 48 hours. The total WOMAC score is between 0 to 96, 0 being the best and 96 being the worst.

Machine Learning

Machine learning (ML) is the process by which a computer program adjusts parameters within the program from actual data, with the aim to detect relations within the data, so that if there is new, not previously seen input, then the system's computer can predict the output. Nowadays, ML is used in several fields such as medicine, marketing, sports, and industry. ML is separated into three categories: supervised learning, unsupervised learning and reinforcement. Supervised learning is the most common method used in machine learning where the dependent or predicted variables (labels) depend on the data. Supervised learning can solve regression and classification problems.

Regression Problem

The regression problem is the study of relations between independent and dependent variables where the dependent variable takes discrete or continuous values as for example, in predicting house sale prices. The regression problem can be solved by several methods, such as the Linear Regression Model, Gradient Boost Tree (GBT), or Decision Tree to approximate the dependent value.

Generalized Linear Model

A Generalized Linear Model (GLM) is a statistical modeling technique formulated by John

Nelder and Robert Wedderburn in 1972. The model can build a linear relationship between the independent and dependent variables, even though the underlying data relationship is not linear. The model is popular for analyzing data since this model can explain the influence of independent variables on a dependent variable. The model consists of three components: the random component, the systematic component, and the link function. The solver of the generalized linear model is used to solve a certain optimization problem with an objective function. Several methods exist for this purpose: Iteratively Reweighted Least Squares (IRLSM), Limited-memory BFGS (LBFGS), or Coordinate Descent.

Support Vector Machine

The Support Vector Machine (SVM) is a supervised learning technique in machine learning for solving classification and regression problems. This technique works best with data of medium size and many attributes. Furthermore, linear and non-linear data can be used by SVM.

Let $T = \{(x_1, y_1), (x_2, y_2), \dots, (x_l, y_l)\}$ be the training data where $x_i \in R^n$ and $y_i \in R$ when $i = 1, 2, \dots, l$. The equation of a SVM is

$$f(x) = (w \cdot x) + b, \quad (1)$$

where $f(x)$ is a predicted function, w is a weight vector and b is a bias value and $b \in R$, $(\cdot)$ is an inner product and x is an input data. Solving the regression problem uses an optimized weight parameter as in the following equation

$$\min_{w, b, \xi^{(*)}} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^l (\xi_i + \xi_i^*) \quad (2)$$

with constrains

1. $y_i - (w \cdot x_i) - b \leq \varepsilon + \xi_i$,
2. $(w \cdot x_i) + b - y_i \leq \varepsilon + \xi_i^*$,

where $\xi_i, \xi_i^*, C \geq 0, \forall i = 1, 2, \dots, l$; ξ_i, ξ_i^* are slack variables; $(\cdot)$ is a scalar product; and C is a penalty parameter. We obtain the dual problem

$$\min_{\alpha, \alpha^*} \frac{1}{2} (\alpha_i^* - \alpha_i)(\alpha_j^* - \alpha_j)(x_i, x_j) + \varepsilon \sum_{i=1}^l (\alpha_i^* - \alpha_i) - \sum_{i=1}^l y_i (\alpha_i^* - \alpha_i), \quad (3)$$

with constrains

$$1. \sum_{i=1}^l (\alpha_i^* - \alpha_i) = 0,$$

$$2. \alpha_i^*, \alpha_i \in [0, C], i = 1, 2, \dots, l,$$

where α_i^* and α_i are Lagrange multipliers.

Kernel

The kernel method is a technique for dealing with non-linear data sets or linearly inseparable data. This method transforms the data into a higher-dimensional space in which a SVM can classify. The kernel method is extremely powerful, and its definition is as follows (Deng *et al.*, 2012):

Definition: A function $K(x, x^*)$ defined on $R^n \times R^n$ is called a kernel on $R^n \times R^n$, or briefly kernel, if there exist a map ϕ from the space R^n to some the Hilbert space

$$\begin{aligned} \phi: R^n &\rightarrow H, \\ x &\mapsto \phi(x), \end{aligned}$$

such that

$$K(x, x^*) = (\phi(x) \cdot \phi(x^*)),$$

where $(\cdot)$ denotes the inner product of the space H . Several types of kernels are typically used, for example, linear kernel, polynomial kernel, Gaussian radial basis function kernel, anova, and epachnenikov. The SVM model with concerned the kernel function is

$$f(x) = \sum_{i=1}^l (\alpha_i^* - \alpha_i) K(x, x_i) + b, \quad (4)$$

where $f(x)$ is a predicted function, α_i^* and α_i are Lagrange multipliers, $b \in R$ is a bias value, $K(x, x_i)$ is a kernel function, and x is an input data.

We obtain the dual problem

$$\begin{aligned} \min_{\alpha, \alpha^*} \frac{1}{2} (\alpha_i^* - \alpha_i)(\alpha_j^* - \alpha_j) k(x_i, x_j) \\ + \varepsilon \sum_{i=1}^l (\alpha_i^* - \alpha_i) - \sum_{i=1}^l y_i (\alpha_i^* - \alpha_i), \end{aligned} \quad (5)$$

with constrains

$$\begin{aligned} 1. \sum_{i=1}^l (\alpha_i^* - \alpha_i) &= 0, \\ 2. \alpha_i^*, \alpha_i &\in [0, C], i = 1, 2, \dots, l, \end{aligned}$$

where α_i^* and α_i are Lagrange multipliers.

Deep Learning

Deep Learning (DL) is a sub-field of machine learning where the algorithms of this technique are

inspired by the construction of neurons in humans, called an *artificial neural network*. The components of an artificial neural network consist of *nodes* (also called neurons), weights between nodes, layers, and activation functions. The output of an artificial neural network is defined by the activation function, for instance a sigmoid function, the hyperbolic tangent function, or a rectifier linear unit.

Gradient Boost Tree

The Gradient Boost Tree (GBT) is a supervised learning technique for solving classification and regression problems. This technique is based on regression and classification trees by creating a model by this technique from an ensemble of weak learning such as a decision tree. The models are built sequentially, and the error of the previous tree model is determined in the next tree model.

GBT is usually used for the reasons that this technique has efficient accuracy, can handle missing data and has high flexibility, since the GBT has several parameters that can be optimized, such as the number of trees (or number of iterations), maximality of depth, loss function and learning rate. The algorithm of GBT follows Friedman's Gradient Boost algorithm. (Natekin and Knoll, 2013).

Friedman's Gradient Boost Algorithm:

Input steps:

- (1) Input the training set

$$T = \{(x_1, y_1), (x_2, y_2), \dots, (x_l, y_l)\},$$

where $x_i \in R^n$, $y_i \in R$, $i = 1, 2, \dots, l$;

- (2) Input the number of iterations M ;
- (3) Choose a differentiable loss function $L(y_i, F(x))$.

Algorithm:

- (1) Initialize the model with a constant value:

$$F_0(x) = \operatorname{argmin}_{\gamma} \sum_{i=1}^l L(y_i, \gamma);$$

- (2) For $m = 1$ to M do
 - (2.1) Compute

$$r_{im} = - \left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)} \right]_{F(x)=F_{m-1}(x)},$$

where $F(x_i)$ is a predicted variable;

- (2.2) Fit a regression tree to the r_{im} value and create terminal regions $R_{j,m}$ for $j = 1, 2, \dots, J_m$;

- (2.3) For $j = 1, 2, \dots, J_m$ compute

$$\gamma_{j,m} = \operatorname{argmin}_{\gamma} \sum_{x_i \in R_{j,m}} L(y_i, F_{m-1}(x_i) + \gamma);$$

- (2.4) Update

$$F_m(x) = F_{m-1}(x) + v \sum_{j=1}^{J_m} \gamma_{j,m} I(x \in R_{j,m})$$

where v and I are learning rate and indicator function, respectively;

- (3) Output $F_m(x)$.

Feature Engineering

Feature engineering (FE) is a technique to consider the attributes used to create a model. This technique has several methodologies such as selecting, extracting, generating or transforming the attributes. Feature engineering is an important technique for machine learning for the reason that it improves accuracy and decreases the complexity of a model. Data scientists often expend a lot of time performing this technique to determine which attributes are important for a model. Feature engineering can be divided into

- 1) feature selection: which features are regarded as important to the model;
- 2) feature extraction to reduce the number of attributes considered by generating new attributes of more significance than the attributes in the original data (Zheng and Casari, 2018; Kokkoti et al., 2020).

Automatic Feature Engineering

In 2015, MIT researchers presented Deep Feature Synthesis (Kanter and Veeramachaneni, 2015), the basic concept behind automatic feature engineering. Deep Feature Synthesis consists of two parts: feature selection and feature generation, where the latter uses the attributes to transform the data using several mathematical operations such as transformation of data with several functions, such as the logarithm function or exponential function, and creating attributes by using the values of some attributes and adding, subtracting or multiplying them together to obtain automatically a new attribute. This technique would be useful because data scientists would otherwise not be able to consider multiple relationships between the data. Nevertheless, this technique considers several forms of attributes with the additional advantage of automatic prediction.

K-Fold Cross Validation

The data set D is divided into K partitions with K-1 partitions used for training and the Kth used for testing. This is repeated K times, with each partition used for testing only once.

Performance

In solving the regression problem, let

$$e_i = \hat{y}_i - y_i, i = 1, \dots, l,$$

where y_i is an observed value, $\hat{y}_i$ is a predicted value and l is the amount of data. In this research study of the regression problem, we consider mainly three measurements: root mean square error, mean absolute deviation, and square error (Botchkarev, 2018).

Root Mean Square Error

Root mean square error (RMSE) is the square root of the sum of the squared errors divided by the number of data points,

$$RMSE = \sqrt{\frac{\sum_{i=1}^l e_i^2}{l}}.$$

Mean Absolute Deviation

Mean absolute deviation (MAD) is the sum of the absolute errors divided by the number of data points,

$$MAD = \frac{\sum_{i=1}^l |e_i|}{l}.$$

Table 1. The attributes used in this research

Attribute	Description
X_1	Sex
X_2	Age
X_3	Body mass index
X_4	Number of months having been able to walk without walking aid.
X_5	Pain when getting in or out of the car
X_6	Pain when lying supine on the bed
X_7	Meeting patient's expectations
X_8	Patient's ability to return to normal activity
X_9	Pain when sitting on chair
X_{10}	Pain when walking on flat surface
X_{11}	Flexibility of the knee after operation
y	Post-surgery WOMAC score

Table 2. Parameters of the Generalized Linear Model

Parameter	Value
Family	Gaussian, Poisson, Tweedie, Negative binomial
Solver	IRLSM, LBFGS, Coordinate descent, Coordinate descent naive
Link function	Identity, Inverse, Logit

Square Error

Square error (SE) is the sum of the squared errors,

$$SE = \sum_{i=1}^l e_i^2.$$

Materials and Methods

Tools

We used the RapidMiner Studio version 9.9 (Education license) on Microsoft Windows 10 to complete this research. RapidMiner is a software tool for data and text mining. It collects all data and handles data, from data preparation to the deployment of predictive models.

Data

The research used the questionnaires collected from the patients' post-operative total knee replacement which Asst. Prof. Lt. Col. Bura Sindhupakorn at Suranaree University of Technology Hospital has collected from 2007 to 2018. The data has 1250 incidences and 21 attributes. In this research, we extracted 12 attributes with dependent variable being the WOMAC score after surgery. (Table 1.)

Optimized Parameters

This step is used to optimize the parameters of the Generalized Linear Model, Support Vector Machine (Philipp *et al.*, 2019), Deep Learning (Koutsoukas *et al.*, 2017), and Gradient Boost Tree (Bentéjac *et al.*, 2020) to set up the parameters in the feature engineering process for the purpose of selecting and generating new attributes that are suitable with each model.

Feature Engineering

This step is used to select and generate attributes by automatic feature engineering (FE) with four techniques including, Generalized Linear Model, Support Vector Machine, Deep Learning, and Gradient Boost Tree. The parameters that were set in GLM were family, solver, and link function. (Table 2.), The parameters that were set in SVM were kernel gamma, kernel sigma1, degree, penalty, epsilon, and kernel. (Table 3.), The parameters that were set in Deep Learning were activation function, loss function, and learning rate. (Table 4.), The parameters that were set in GBT were the number of trees, maximal depth, and learning rate. (Table 5.)

Model creation

In this step we created four models by the Gradient Boost Tree; each model employed different attributes and each attribute obtained from

feature engineering. The validation of each model used K -fold cross-validation with $K = 10$. Table 5 shows the parameters used to optimize each model (Bentéjac *et al*, 2020).

Results and Discussion

Data Set

The data set has 1,250 incidences and 12 attributes as shown in Table 6.

Novel Attributes

This part describes the selected attributes and the new attributes generated by automatic feature engineering for GLM, SVM, DL, and GBT. The following novel attributes were obtained.

Generalized Linear Model

The parameters of the generalized linear model were set using automatic feature engineering: the family was Gaussian; link function was the identity; and solver was IRLSM. The group attribute (FE+GLM) was obtained from automatic feature engineering on the Generalized Linear Model such as X_3 , X_4 , X_7 , X_8 and G_1 , where G_1 is the new attribute, namely the value of the logarithm of the maximum value between X_4 and X_7 (i.e, $G_1 = \log(\max\{X_4, X_7\})$).

Support Vector Machine

The parameters of the Support Vector Machine were set with the automatic feature engineering: the kernel was radial, the kernel gamma was 3, the penalty was 38 and epsilon was 2. The group

attribute (FE+SVM) was obtained from automatic feature engineering on the Support Vector Machine including X_2 , X_3 , X_4 , X_7 , X_8 and X_{11} .

Deep Learning

The parameters of Deep Learning were set by automatic feature engineering: the activation function was rectifier, the loss function was quadratic, and the learning rate was 0.01. The group attribute (FE+DL) was obtained from automatic feature engineering on Deep Learning including X_2 , X_3 , X_4 , X_5 , X_7 , X_8 , X_{11} and G_2 , where G_2 was the new attribute mean, the value of X_6 divided by value of X_7 .

Gradient Boost Tree

The parameters of the Gradient Boost Tree were set by automatic feature engineering: the number of trees was 200, maximal depth was 11 and the learning rate was 0.1. The group attribute (FE+GBT) was obtained from automatic feature engineering on the Gradient Boost Tree including X_2 , X_3 , X_5 , X_7 , X_8 , X_{10} , X_{11} , G_3 and G_4 , where G_3 and G_4 were the new attributes, namely, the value of X_8 minus by value of X_{11} , and value of X_3 multiplied by value of X_4 , respectively.

Table 6. Attributes employed

Attribute	Type	Description
X_1	Independent	1: Male, 2: Female
X_2	Independent	Integer (47-85)
X_3	Independent	Real numbers (18.91-41.02)
X_4	Independent	Real numbers (0.46-24)
X_5	Independent	1: Uncertain 2: Unsatisfied 3: Satisfied 4: Very satisfied
X_6	Independent	1: Uncertain 2: Unsatisfied 3: Satisfied 4: Very satisfied
X_7	Independent	1: Uncertain 2: Unsatisfied 3: Satisfied 4: Very satisfied
X_8	Independent	1: Uncertain 2: Unsatisfied 3: Satisfied 4: Very satisfied
X_9	Independent	1: Uncertain 2: Unsatisfied 3: Satisfied 4: Very satisfied
X_{10}	Independent	1: Uncertain 2: Unsatisfied 3: Satisfied 4: Very satisfied
X_{11}	Independent	1: 0°-45° 2: 46°-90° 3: 91°-135° 4: more than 135°
y	Dependent	Score (15.2 – 100)

Table 3. Parameters of the Support Vector Machine

Parameter	Value
Kernel gamma	0, 1, 2, ..., 50
Kernel sigma1	0, 1, 2, 3, 4, 5
Degree	0, 1, 2, ..., 10
Penalty	-1, 0, 1, ..., 100
Epsilon	0.0001, 0.001, 0.01, 0.1, 0, 1, 2, 3
Kernel	dot, radial, anova, epachnenikov

Table 4. Parameters of Deep Learning

Parameter	Value
Activation function	Rectifier, ExpRectifier, Tanh, Maxout
Loss function	Absolute, Quadratic, Quantile
Learning rate	10^{-7} , 10^{-6} , 10^{-5} , ..., 10^{-1}

Table 5. Parameters of Deep Learning

Parameter	Value
Number of trees (NT)	100, 200, 300, 400, 500
Maximal depth (MD)	1, 2, 3, ..., 30
Learning rate (LR)	0.0001, 0.001, 0.01, 0.1

Table 7. RMSE, MAD and SE of the Models

Model	Attributes	NT	MD	LR	RMSE	MAD	SE
GBT	FE+GLM	200	15	0.1	5.819±0.801	3.738±0.481	34.443±9.233
GBT	FE+SVM	200	10	0.1	5.517±0.699	3.660±0.442	30.880±7.781
GBT	FE+DL	200	11	0.1	5.544±0.652	3.632±0.414	31.113±7.288
GBT	FE+GBT	200	11	0.1	5.316±0.539	3.529±0.370	28.525±5.828

Table 8. Parameters of Gradient Boost Tree

Parameter	Value
Number of trees	100, 200, 300, 400, 500
Learning rate	0.0001, 0.001, 0.01, 0.1

Table 9. Parameters of Gradient Boost Tree

Model	Attributes	NT	MD	LR	RMSE	MAD	SE
GBT	FE+GBT	500	11	0.01	5.459±0.654	3.678±0.427	30.190±7.289
GBT	FE+GBT	500	11	0.02	5.366±0.563	3.594±0.379	29.075±6.212
GBT	FE+GBT	500	11	0.03	5.334±0.544	3.549±0.367	28.713±5.934
GBT	FE+GBT	400	11	0.04	5.311±0.538	3.550±0.376	28.472±5.881
GBT	FE+GBT	400	11	0.05	5.331±0.517	3.536±0.365	28.659±5.606
GBT	FE+GBT	200	11	0.06	5.334±0.566	3.561±0.392	28.743±6.199
GBT	FE+GBT	200	11	0.07	5.312±0.565	3.532±0.391	28.501±6.177
GBT	FE+GBT	200	11	0.08	5.317±0.544	3.540±0.352	28.533±5.922
GBT	FE+GBT	200	11	0.09	5.338±0.508	3.543±0.356	28.724±5.552
GBT	FE+GBT	200	11	0.1	5.316±0.539	3.529±0.370	28.525±5.828

Evaluation of model

This part demonstrates the efficiency of the Gradient Boost Tree with other attributes where 10-fold cross-validation is used to validate the model. The model can be evaluated by RMSE, MAD, and SE metrics, keeping in mind that a lower value of each metric is better than a high value. The values of these metrics are shown in Table 7.

Performance of learning rate of Gradient Boost Tree

From Table 8, we can see that the best model is the Gradient Boost Tree with attributes originating by automatic feature engineering on Gradient Boost Tree (or group attributes 4), so only the learning rate parameter is taken into account. As a result, the model's parameters remain unchanged; only the learning rate parameter is changed.

Figure 1 shows the RMSE for various learning rate values in the range of 0, 0.1, 0.2, ..., 1. The RMSE is lowest when the learning rate is 0.1. After that, the model is tweaked by changing two parameters: the learning rate and the number of trees. The parameters chosen were as listed in Table 8.

The results shown in Figure 2 demonstrate that for all choices of the number of trees, a learning rate of 0.1 is better than the other learning rates. Following that, more fine-tuning of the learning rate and number of trees produced the data shown in Table 9, showing that the learning rate of 0.07 and the number of trees of 200 have an RMSE of 5.312 ± 0.565 . The RMSE for a learning rate of 0.04 and a number of trees of 400 is 5.311 ± 0.538 . As a result, both models in

Table 9 have a lower RMSE than the other models in Table 8. This demonstrates that the learning rate is a factor in the gradient boosted tree's performance.

Conclusions

A model has been created in which different group attributes were measured by RMSE, MAD, and SE in Table 8. The result of the table shows that the Gradient Boost Tree using group 4 attributes received from automatic feature engineering on the Gradient Boost Tree is the best, with second, third, and fourth-best performance being attributes from automatic feature engineering on Deep Learning, Support Vector Machine, and the Generalized Linear Model respectively. Figure 1, Figure 2, and Table 9, show that the learning rate of the Gradient Boost Tree affects the efficiency of the model.

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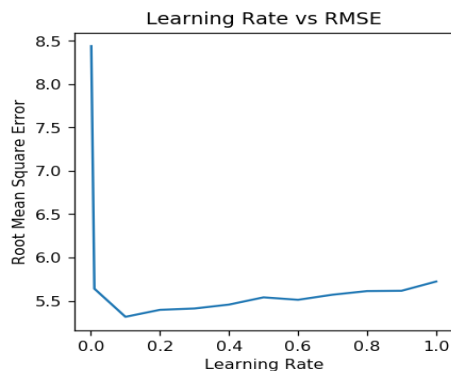


Figure 1. Tune Learning Rate in GBT

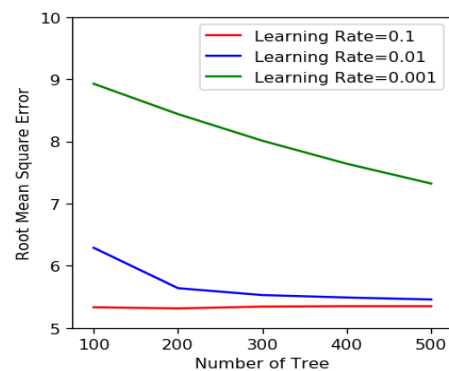


Figure 2. Tune Learning Rate and Number of Trees in GBT

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